Segmentation by Humans - Gestalt Principles

1. **Proximity Principle** - Objects that are closer to each other are grouped together
2. **Similarity Principle** - Similar objects are grouped together. Proximity principle is given more priority than similarity principle
3. **Commom Fate** - Objects with similar motion or change in appearence are grouped together
4. **Common region/Connectivity** - Connected objects are grouped together
5. **Continuity Principle** - Features on a cintinuous curve are grouped together.
6. **Symmetry Principle** - Parallel and symmetrical features are grouped together
7. **Illusory Contours - Illusion's grouping** - Illusory or subjective contours are percieved.

Two approaches towards Segmentation

1. **Top-Down Segmentation (Model - Driven)** - It uses high-level knowledge, such as object shapes. Pixels belong together as they come from the same object.

Methods:

- **Active contours (Snake)** - Segment using curves and not by pixels. It is used in medical imaging, industrial inspection, object tracking etc.

> $V(S) = (X'(S)Y(S))$ $E = E_{internal} + E_{image} + E_{external}$

- **Scissors** - Intelligent Scissors/Live-Wire is an interactive image segmentation technique. The algorithm finds optimal (shortest) boundary path b/w clicks. It is used in photo editing, medical images etc.

2. **Bottom-up Segmentation (Data driven)** - Group pixels based on local cues like color or textue. Pixels belong together as they look similar.

Methods:

- **Thresholding**
- **Region growing**

K- Means Clustering

Choice of K : k = 2 - background or foreground k = 5 - coarse regions k = 50 - fine over segmentation

Disadvantages of k means:

- We have to pick optimal value of k
- It maybe sensitive to initialization and outliers

Mean Shift Segmentation

Choosing window size: Small bandwidth - many small clusters Large bandwidth - few large clusters

- Each hill represents a cluster.

- Simple but computationally expensive
- Clustering depends on window size w
- Robust to outliers
- Clustering depends on window size w

Example: Data Points [1,2,3,7,8,9,10,11] **Solution:** Each point is a potential cluster center Mean Shift Iteration (for point 1):

- Window: 2, radius = 2
- Mean Shift: (1 + 2 + 3)/ 3 = 2 New center is 2
- Window: Points w bandwidth 2 of 2 : [1,2,3]
- Mean Shift : (1 + 2 + 3)/ 3 = 2 1 is the hill with 2 as peak

Mean Shift Iteration (for point 7):

- Window: Points w bandwidth 2 of 7 : [7,8,9]
- Mean Shift : 7 + 8 + 9/3 = 8, center = 8
- Window: [7,8,9,10]
- Mean Shift: 7 + 8 + 9 + 10/4 = 8.5
- Window: Points w bandwidth 2 of 8.5 is [7,8,9,10]
- Converges at 8.5 7 is the hill with 8.5 as peak

Graph Based Segmentation

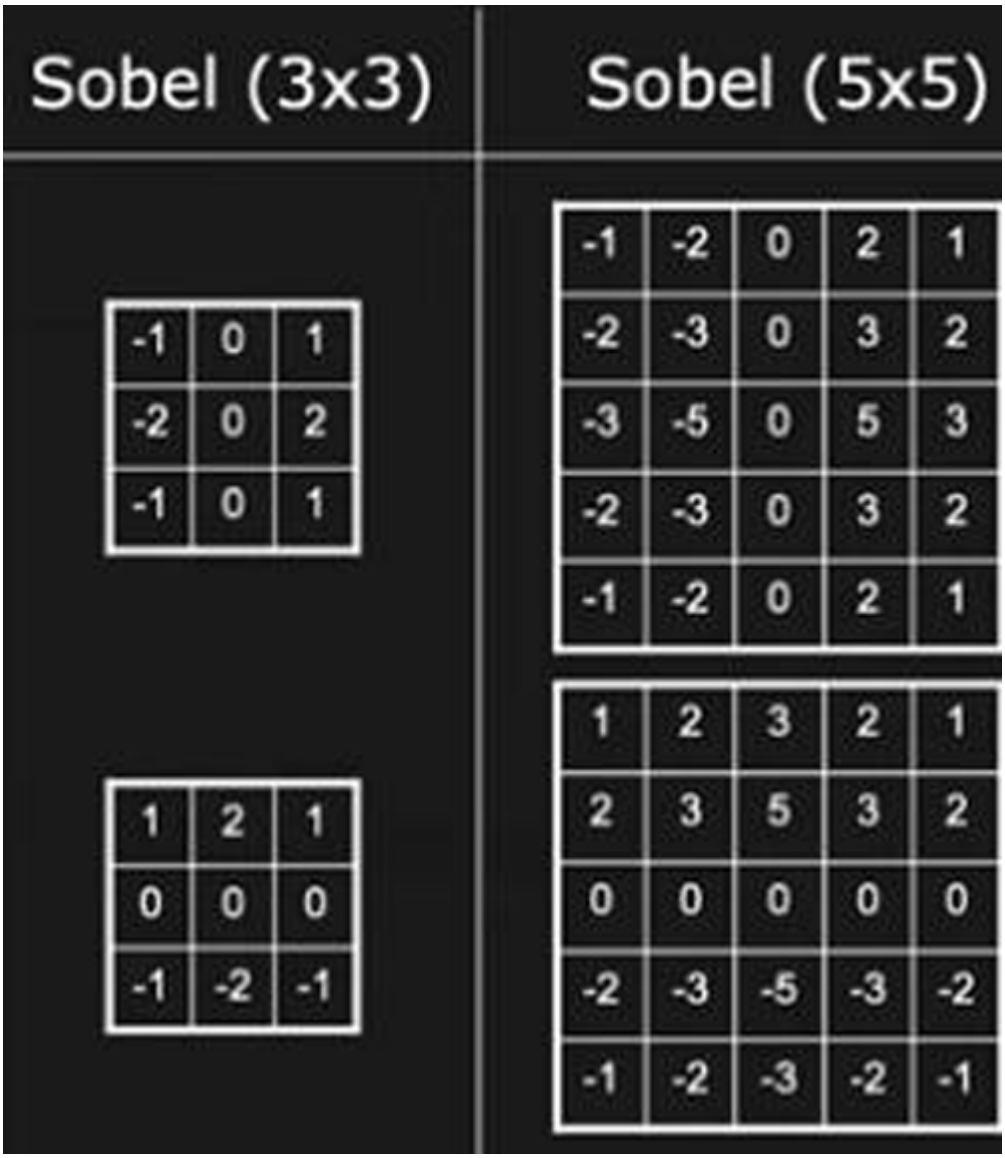
- Every pixel is a vertex
- An edge btwn each pair of pixels (vertices)

> G = (V,E)

- Measuring affinity: smaller the dissimilarity, larger the affinity
- Normalized Cut - Minimize cost of Normalized Cut during partition.. The Ncut value ranges from 0 to 2. 0 indicates high-quality cut and 2 indicates poor partiotion. **Example** A = {1}, B = {2,3,4} Cut (A,B) = (1,2) = 5; (1,3) = 1; (2,3) = 1; (2,4) = 1; (3,4) = 5 Assoc(A,V) = (1,2) + (1,3) = 6; Ncut = 6/6 + 6/7 = Approches towards 2

Edge Based Segmentation

- It is a method of segmenting an image into regions of discontinuity
- It detects boundaries using operators like Sobel, Canny, Laplacian
- Converts 2D image into a set of points.
- Gaussian - Computes first order derivations like sobel
- Gradient - Cpmutes second-order derivations like Canny



Question $$ I =\begin{bmatrix} 100 & 120 & 130 & 140 \\ 90&110 &120 &130\\ 80&100 &110 &120 \\ 70&90 &100 &110 \\ \end{bmatrix} $$

Compute gradients and orientation for all pixels. Use 0 padding and sobel opertion for edge detection and orientation : $$ sobel matrix_x =\begin{bmatrix}
-1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} $$

$$ sobel matrix_y =\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} $$

Solution

Add zeros in all 4 rows and coloums and multiply sobel x matrix with 3 X 3 I matrix

$$ I_x=\begin{bmatrix} 350& 90 &60 & -380 \\ 440 & 120 & 80 &-480 \\ 400&120 &80 &-440 \\ 280& 90 & 60 & -310 \\ \end{bmatrix} $$

$$ I_y=\begin{bmatrix} -290& -430 &-480 & -380 \\ 60 & 80 & 80 &60 \\ 60&80 &80 &60 \\ 260& 390 & 440 & 350 \\ \end{bmatrix} $$

Now, perform sqrt of I_x + I_y

$$ Magnitude matrix =\begin{bmatrix} 440& 439 &483 &537 \\ 444&144 &113 & 483\\ 404& 144 &113 &444 \\ 382&400 & 444 & 487\\ \end{bmatrix} $$

Orientation matrix = tan (theta) (I_x / I_y)

Feature Detection and Matching

Image Stitching: Aligning two images so that they can be seamlessly stitched into a complete mosiac.

Object Recognition: Identifying specific objects within an image by comparing extracted features to a known database of images.

Image Registration: Process of aligning two or more images of the same taken at different times, angles or sensors. It is used in medical imaging, computer vision and robotics.

3D Construction: Eshtablish a dense set of correspondences so that a 3D model can be generated.

Object Tracking: Feature detectiona nd matching is used to identify and locate key points on an object across multiple frames.

Feature

A piece of info relevant for solving the computational task

Main Categories:

1. **Key Point/Interest Point/Corner Feature:** Peaks, corners, appearance of patches of pixels surrounding the point location.

2. **Egdes(object boundaries):** It is based on orientation and local appearance

Main components of Feature Detection and Matching

1. **Detection** Identifying intrest point
2. **Description** It gives descriptor vector for each intrest point
3. **Matching** Descriptors are compared across the images to identify similar images. It gives a set of pairs of matching images.

SIFT Detector

- Scale Invarient Feature Transfrom (SIFT)
- It is close ot real time performance
- Actual size considered in SIFT IS 16 X 16

Main steps involved

1. Detecting Intrest points: edges, corners, peaks
2. Extract feature descriptor

Attributes of Key / Interest Points

- Well defined representation, position in image
- Should be invarient to image rotation and camera lighting
- Blobs: A region of an image that differs from its surrounding area in color or brightness

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Scale Invarience

- Once detected the interest points, normalize/scale up or down to match the other for the applications
- Increase or decrase the size as per sigma value

Rotation Invariant

- Construct gradient magnitude and orientation for each blob region.
- Build an orientation histogram and assign dominant direction.

Application of SIFT

- **Image registration or alignment(2D)** : Process of transforming multiple images into a common coordinate system
- **Object recognition** : Help identify objects by finding distinctive features
- **Image Stitching** : Used to stitch images together for panoramic photography

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